

MSML 604 – Introduction to Optimization

Self-Defined Optimization Project: Instructions document

1 Project Overview

The goal of the project is to provide you with an opportunity to apply concepts from **optimization theory** to a real-world or research-oriented problem.

You are strongly encouraged to review **recent research literature** related to your chosen topic and, where possible, implement existing methods and propose improvements, extensions, or comparative analyses grounded in optimization principles.

2 Project Objectives

The objective of this project is to apply optimization principles to a well-defined problem and to demonstrate a strong understanding of optimization-based modeling and analysis. Each project should clearly formulate the problem within a rigorous optimization framework, implement an appropriate solution strategy, and assess the results through empirical evaluation with thoughtful interpretation.

Given the limited timeframe, preliminary findings that persuasively support the proposed approach are acceptable. **Because convex optimization is a primary focus of this course and provides substantial theoretical and computational advantages, students are expected to first investigate convex optimization methods before turning to alternative techniques.** This process, including any comparisons, extensions, or methodological choices, should be clearly documented in the project report. A pure literature review is discouraged; however, it will be considered if it includes original insights, novel interpretations, or meaningful extensions.

3 Example Project Ideas

The following examples are provided only for guidance. You are free to explore your own ideas.

- **Portfolio Optimization:** Formulate and solve a mean-variance portfolio optimization problem under realistic constraints such as budget, risk limits, or transaction costs.
- **Resource Allocation Optimization:** Model a constrained allocation problem (e.g., logistics, scheduling, or energy distribution) and solve it using optimization techniques.
- **Regularized Regression Analysis:** Implement and compare Lasso and Ridge regression from an optimization perspective. Analyze sparsity, solution paths, and convergence behavior using convex optimization solvers.
- **ML Optimization algorithms:** Compare different ML training algorithms and study their performance; Visualize loss landscape of Neural nets

- **Multiobjective Optimization in ML systems:** Optimize energy consumption and performance of ML models. This is particularly useful for edge AI applications.
- **Bilevel Optimization in ML systems:** Regularization tuning, Optimizing ML architecture
- **Distributed Optimization:** Communication vs Computation tradeoffs in a distributed setting
- **Optimizing ML Collective Communications:** Optimize collective communication for training and inferencing on a network of GPUs
- More details to follow

4 Group Formation

You may work individually or in **self-defined groups of up to three (3) members**.

A Google Form will be distributed by the TA, to collect project registration details. **Only one student per group should submit the form.** The student submitting should fill up the details of all the groups members along with project name and a short description.

Important: The student submitting the form must include the details of *all* group members. Individual submissions from multiple members of the same group should be avoided.

Registration Deadline: *March 11, 2026*

5 Project Checkpoint (Abstract Submission)

Checkpoint Deadline: *April 6, 2026*

Each group must submit a **1–2 page project abstract** outlining the problem being addressed, its motivation and application context, and the optimization formulation you plan to study. The abstract should also describe the proposed methodology, the overall goals of the project, and the outcomes you aim to achieve.

6 Final Deliverables

Final Submission Deadline: *May 4, 2026*

Each group must submit the following:

6.1 Source Code (Required)

- Complete end-to-end implementation hosted on a **GitHub repository**.
- Code should be organized, documented, and reproducible.

6.2 Final Report (Required)

The report should follow a **research paper-style format** and include:

- Problem definition and motivation
- Methodology and optimization approach
- Experiments and results
- Discussion and conclusions
- Proper references and citations (mandatory)

Note: Source code should not be included within the report, although pseudocode or algorithmic descriptions may be included if helpful.

Figures, diagrams, and charts are strongly encouraged. All visuals must be clearly labeled and properly captioned.

6.3 Presentation Slides (Required)

- Have about 10-15 slides
- Slides should clearly communicate problem setup, methodology, results, and conclusions.

6.4 Demo (Optional)

A live or recorded demo is **not required**, but projects that include demonstrations of working implementations will be viewed favorably.

7 General Expectations

- Projects should demonstrate clear use of optimization concepts and techniques.
- Proper citation of external resources and research papers is required.
- Collaboration must follow university academic integrity policies.
- Projects will be evaluated based on clarity, correctness, depth of analysis, and understanding of optimization principles.
- Gen AI tools (e.g., ChatGPT, Claude, etc.) can be used for the project. However, code should not be directly copied.

Additional details regarding submission logistics and grading rubrics will be provided separately.